Network Intrusion Detection & Prevention Systems Exploitation

Use of Suricata IDS

(Part 1)

I. Installing and configuring Suricata on CentOS 8

1. Start by installing Suricata from OISF (Open Information Security Foundation, A non-profit foundation organized to build a next generation IDS/IPS engine) provided RPMs

$ sudo sed -i 's|mirrorlist=|#mirrorlist=|g' /etc/yum.repos.d/*.repo

```
[centos@centosstream8 ~]$ sudo sed -i 's|mirrorlist=|#mirrorlist=|g' /etc/yu
m.repos.d/*.repo
```

$ sudo sed -i 's|#baseurl=http://mirror.centos.org|baseurl=https://vault.centos.org|g' /etc/yum.repos.d/*.repo

```
[centos@centosstream8 ~]$ sudo sed -i 's|#baseurl=http://mirror.centos.org|b
aseurl=https://vault.centos.org|g' /etc/yum.repos.d/*.repo
```

$ sudo dnf clean all && sudo dnf makecache

```
[centos@centosstream8 ~]$ sudo dnf clean all && sudo dnf makecache
0 files removed
CentOS Stream 8 - AppStream                   1.4 MB/s |  29 MB     00:20
CentOS Stream 8 - BaseOS                      648 kB/s |  10 MB     00:16
CentOS Stream 8 - Extras                       16 kB/s |  18 kB     00:01
Metadata cache created.
```

$ sudo yum install epel-release yum-plugin-copr

```
[centos@centosstream8 ~]$ sudo yum install epel-release yum-plugin-copr
Last metadata expiration check: 0:03:18 ago on Mon 13 Oct 2025 02:33:08 PM I
ST.
Package dnf-plugins-core-4.0.21-3.el8.noarch is already installed.
Dependencies resolved.
```

```
  libsolv-0.7.20-6.el8.x86_64
  python3-dnf-4.7.0-20.el8.noarch
  python3-dnf-plugins-core-4.0.21-25.el8.noarch
  python3-hawkey-0.63.0-19.el8.x86_64
  python3-libdnf-0.63.0-19.el8.x86_64
  yum-4.7.0-20.el8.noarch
Installed:
  epel-release-8-11.el8.noarch

Complete!
```

$ sudo yum copr enable @oisf/suricata-6.0

```
[centos@centosstream8 ~]$ sudo yum copr enable @oisf/suricata-6.0
Enabling a Copr repository. Please note that this repository is not part
of the main distribution, and quality may vary.

The Fedora Project does not exercise any power over the contents of
this repository beyond the rules outlined in the Copr FAQ at
<https://docs.pagure.org/copr.copr/user_documentation.html#what-i-can-build-
in-copr>,
and packages are not held to any quality or security level.

Please do not file bug reports about these packages in Fedora
Bugzilla. In case of problems, contact the owner of this repository.

Do you really want to enable copr.fedorainfracloud.org/@oisf/suricata-6.0? [
y/N]: y
Repository successfully enabled.
```

$ sudo rpm -Uvh https://rpmfind.net/linux/epel/8/Everything/x86_64/Packages/h/hiredis-0.13.3-13.el8.x86_64.rpm

```
[centos@centosstream8 ~]$ sudo rpm -Uvh https://rpmfind.net/linux/epel/8/Eve
rything/x86_64/Packages/h/hiredis-0.13.3-13.el8.x86_64.rpm
Retrieving https://rpmfind.net/linux/epel/8/Everything/x86_64/Packages/h/hir
edis-0.13.3-13.el8.x86_64.rpm
warning: /var/tmp/rpm-tmp.vAWiAX: Header V3 RSA/SHA256 Signature, key ID 2f8
6d6a1: NOKEY
Verifying...                                                            (100
################################## [100%]
Preparing...                                                            (100
################################## [100%]
Updating / installing...
   1:hiredis-0.13.3-13.el8                                              ( 43
################################## [100%]
```

$ sudo rpm -Uvh http://rpmfind.net/linux/epel/8/Everything/x86_64/Packages/h/hyperscan-5.3.0-5.el8.x86_64.rpm

```
[centos@centosstream8 ~]$ sudo rpm -Uvh http://rpmfind.net/linux/epel/8/Ever
ything/x86_64/Packages/h/hyperscan-5.3.0-5.el8.x86_64.rpm
Retrieving http://rpmfind.net/linux/epel/8/Everything/x86_64/Packages/h/hype
rscan-5.3.0-5.el8.x86_64.rpm
warning: /var/tmp/rpm-tmp.VvYSJK: Header V4 RSA/SHA256 Signature, key ID 2f8
6d6a1: NOKEY
Verifying...                                                            (100
################################## [100%]
Preparing...                                                            (100
################################## [100%]
Updating / installing...
   1:hyperscan-5.3.0-5.el8                                              (  2
################################## [100%]
```

$ sudo yum install suricata

```
[centos@centosstream8 ~]$ sudo yum install suricata
Last metadata expiration check: 0:00:04 ago on Mon 13 Oct 2025 03:15:19 PM I
ST.
Dependencies resolved.
=============================================================================
 Package            Arch   Version        Repository                     Size
=============================================================================
Installing:
 suricata           x86_64 1:6.0.20-1.el8 copr:copr.fedorainfracloud.org:gro
up_oisf:suricata-6.0
                                                                        2.2 M
Upgrading:
 jansson            x86_64 2.14-1.el8     baseos                         47 k
Installing dependencies:
 libnetfilter_queue x86_64 1.0.4-3.el8    baseos                         31 k
```

```
Upgraded:
  jansson-2.14-1.el8.x86_64
Installed:
  libnetfilter_queue-1.0.4-3.el8.x86_64    suricata-1:6.0.20-1.el8.x86_64

Complete!
```

2. Check that the following files/directories are now available

- /etc/suricata/reference.config: a file defining all the reference types (url, CVE-database, ...)

- /etc/suricata/classification.config: a file defining the classification of rules and alerts and assign to each class type a priority

- /etc/suricata/suricata.yaml: Suricata main configuration file

```
[centos@centosstream8 ~]$ sudo ls -la /etc/suricata/
total 104
drwxr-x---.   3 suricata suricata   117 Oct 13 15:15 .
drwxr-xr-x. 143 root     root      8192 Oct 13 15:16 ..
-rw-r-----.   1 suricata suricata  3327 Jun 27  2024 classification.config
-rw-r-----.   1 suricata suricata  1375 Jun 27  2024 reference.config
drwxr-x---.   2 suricata suricata  4096 Oct 13 15:15 rules
-rw-r-----.   1 suricata suricata 76879 Jun 27  2024 suricata.yaml
-rw-r-----.   1 suricata suricata  1644 Jun 27  2024 threshold.config
```

- /var/log/suricata/: a directory for Suricata's log information

```
[centos@centosstream8 ~]$ sudo ls -la /var/log/suricata/
total 4
drwxr-x---.  2 suricata suricata    6 Jun 27  2024 .
drwxr-xr-x. 17 root     root     4096 Oct 13 15:15 ..
```

3. Before deployment, Suricata must be configured for the following variables:

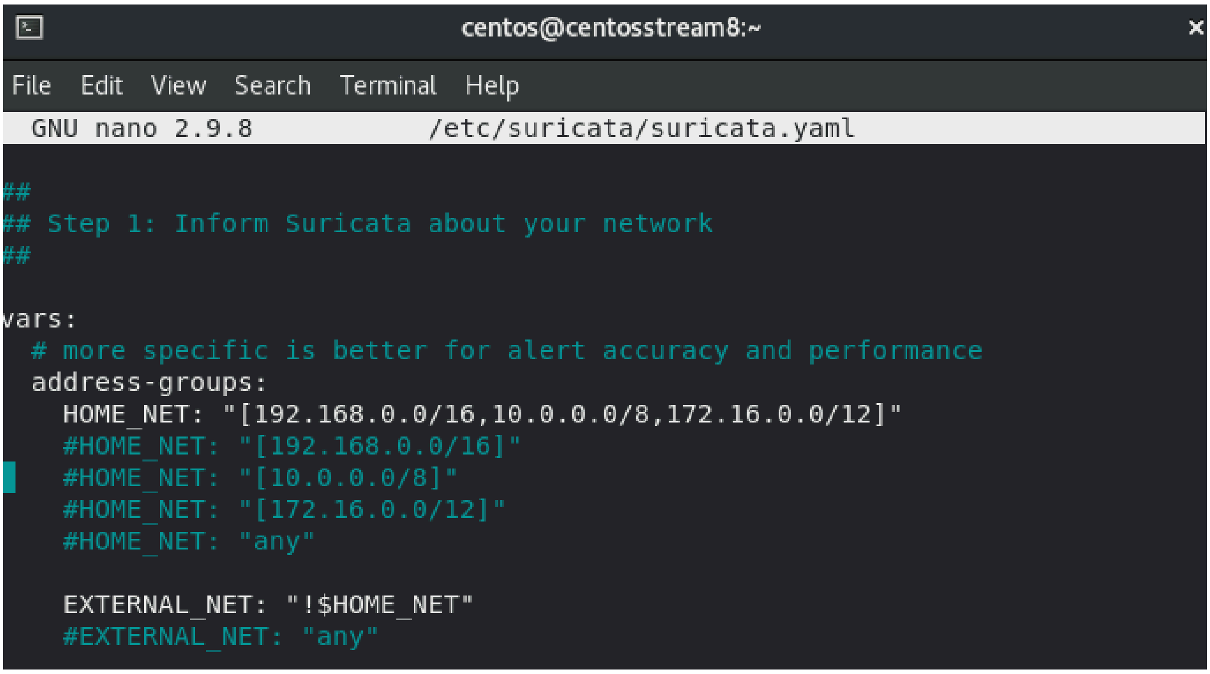
The network interface to use, The IP range of the internal network, and the IP range of the external network.

- HOME_NET : used to define the internal network that is to be protected;

- EXTERNAL_NET : used to define the external network;

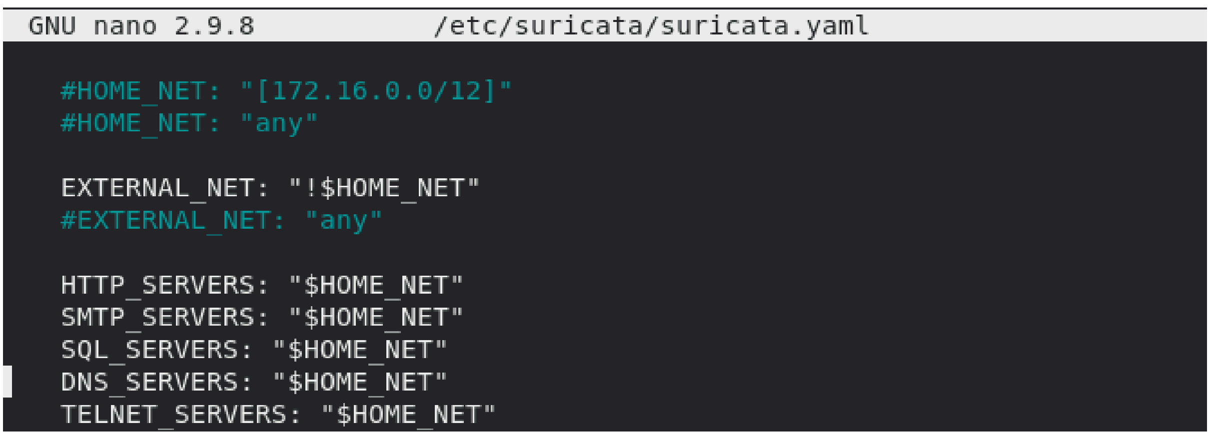
4. In the configuration file /etc/suricata/suricata.yaml

check that the IP address of your network is already defined in the variable $HOME_NET, and that the variable EXTERNAL_NET is set to !$HOME_NET. This way, every IP address excepting the set of HOME_NET will be considered as external



5. Check also that the default values of the variables

HTTP_SERVERS, SMTP_SERVERS, SQL_SERVERS, DNS_SERVERS and TELNET_SERVERS, are set to HOME_NET (i.e., the supervised SMTP, SQL, DNS, and TELNET servers are located on the internal network)



6. Under the af-packet section, set the value of the interface to your interface name.

```
 GNU nano 2.9.8              /etc/suricata/suricata.yaml                Modified

## and PF_RING.
##

# Linux high speed capture support
af-packet:
  - interface: enp0s3
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
```

7. In /etc/suricata/suricata.yaml verify that your default-rule-path and rule-files are configured as follows.

```
 GNU nano 2.9.8              /etc/suricata/suricata.yaml                Modified

## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
  - suricata.rules
```

8. Disable Suricata packet offloading by disabling interface Large Receive Offload (LRO)/Generic Receive Offload (GRO)

$ sudo ethtool -K enp0s3 gro off lro off

```
[centos@centosstream8 ~]$ sudo ethtool -K enp0s3 gro off lro off
```

9. Check now if these features were disabled

$ sudo ethtool -k enp0s3 | grep -iE "generic|large"

```
[centos@centosstream8 ~]$ sudo ethtool -k enp0s3 | grep -iE "generic|large"
        tx-checksum-ip-generic: on
generic-segmentation-offload: on
generic-receive-offload: off
large-receive-offload: off [fixed]
```

10. To get the Emerging Threats Open ruleset you can run the following command which downloads and installs the best version (of the ruleset) for the version of Suricata found and outputs the full set into suricata.rules.

The same command is also used to update the detection rules

sudo suricata-update

```
[centos@centosstream8 ~]$ sudo suricata-update
13/10/2025 -- 16:23:26 - <Info> -- Using data-directory /var/lib/suricata.
13/10/2025 -- 16:23:26 - <Info> -- Using Suricata configuration /etc/suricat
a/suricata.yaml
13/10/2025 -- 16:23:26 - <Info> -- Using /usr/share/suricata/rules for Suric
ata provided rules.
13/10/2025 -- 16:23:26 - <Info> -- Found Suricata version 6.0.20 at /sbin/su
ricata.
13/10/2025 -- 16:23:26 - <Info> -- Loading /etc/suricata/suricata.yaml
13/10/2025 -- 16:23:26 - <Info> -- Disabling rules for protocol http2
13/10/2025 -- 16:23:26 - <Info> -- Disabling rules for protocol modbus
13/10/2025 -- 16:23:26 - <Info> -- Disabling rules for protocol dnp3
13/10/2025 -- 16:23:26 - <Info> -- Disabling rules for protocol enip
13/10/2025 -- 16:23:26 - <Info> -- No sources configured, will use Emerging
Threats Open
13/10/2025 -- 16:23:26 - <Info> -- Fetching https://rules.emergingthreats.ne
t/open/suricata-6.0.20/emerging.rules.tar.gz.
  20% - 1056768/5136868
```

11. Setup the network interface that Suricata listens on, in the file /etc/sysconfig/suricata.

The default version of this file specifies the eth0 interface, but you'll need to change this to enp0s3 (the interface name can be determined using ifconfig command) in OPTIONS line in /etc/sysconfig/suricata to look like:

OPTIONS="-i enp0s3 --user suricata "

```
  GNU nano 2.9.8              /etc/sysconfig/suricata              Modified

# The following parameters are the most commonly needed to configure
# suricata. A full list can be seen by running /sbin/suricata --help
# -i <network interface device>
# --user <acct name>
# --group <group name>

# Add options to be passed to the daemon
OPTIONS="-i enp0s3 --user suricata "
```

12. Verify the version of the installed Suricata

suricata -V

```
[centos@centosstream8 ~]$ suricata -V
This is Suricata version 6.0.20 RELEASE
```

13. To test the Suricata rules for syntax errors, especially when you create your own custom rules, use the command:

suricata -c /etc/suricata/suricata.yaml -T -v

```
[centos@centosstream8 ~]$ sudo suricata -c /etc/suricata/suricata.yaml -T -v

13/10/2025 -- 16:28:18 - <Info> - Running suricata under test mode
13/10/2025 -- 16:28:19 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYSTEM mode
13/10/2025 -- 16:28:19 - <Info> - CPUs/cores online: 2
13/10/2025 -- 16:28:19 - <Info> - Setting engine mode to IDS mode by default
13/10/2025 -- 16:28:19 - <Info> - master exception-policy set to: auto
13/10/2025 -- 16:28:19 - <Info> - fast output device (regular) initialized: fast.log
```

```
13/10/2025 -- 16:29:06 - <Notice> - Configuration provided was successfully loaded. Exiting.
13/10/2025 -- 16:29:07 - <Info> - cleaning up signature grouping structure... complete
```

14. Start suricata using the folloing command, and leave the terminal open:

suricata -c /etc/suricata/suricata.yaml -v -i enp0s3

Terminal 1 :

```
[centos@centosstream8 ~]$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
13/10/2025 -- 16:34:50 - <Notice> - This is Suricata version 6.0.20 RELEASE running in SYSTEM mode
13/10/2025 -- 16:34:50 - <Info> - CPUs/cores online: 2
13/10/2025 -- 16:34:50 - <Info> - Setting engine mode to IDS mode by default
13/10/2025 -- 16:34:50 - <Info> - master exception-policy set to: auto
13/10/2025 -- 16:34:50 - <Info> - Found an MTU of 1500 for 'enp0s3'
13/10/2025 -- 16:34:50 - <Info> - Found an MTU of 1500 for 'enp0s3'
13/10/2025 -- 16:34:50 - <Info> - fast output device (regular) initialized: fast.log
```

15. To make sure that no error has occurred when suricata has started, you can check the Suricata log:

tail -f /var/log/suricata/suricata.log

Terminal 2 :

```
[centos@centosstream8 ~]$ sudo tail -f /var/log/suricata/suricata.log
13/10/2025 -- 16:34:50 - <Info> - stats output device (regular) initialized: stats.log
13/10/2025 -- 16:34:50 - <Info> - Running in live mode, activating unix socket
13/10/2025 -- 16:35:23 - <Info> - 1 rule files processed. 45700 rules successfully loaded, 0 rules failed
13/10/2025 -- 16:35:23 - <Info> - Threshold config parsed: 0 rule(s) found
13/10/2025 -- 16:35:24 - <Info> - 45703 signatures processed. 964 are IP-only rules, 5141 are inspecting packet payload, 39379 inspect application layer, 108 are decoder event only
13/10/2025 -- 16:35:42 - <Info> - Going to use 2 thread(s)
13/10/2025 -- 16:35:42 - <Info> - Running in live mode, activating unix socket
13/10/2025 -- 16:35:42 - <Info> - Using unix socket file '/var/run/suricata/suricata-command.socket'
```

II. Running Suricata in IDS mode

The aim of this section is to run Suricata in passive IDS mode.

16. Many rules are enabled by the default installation within the file /var/lib/suricata/rules/suricata.rules

Count the number of these detection rules using the command:

wc -l /var/lib/suricata/rules/suricata.rules

Terminal 3 :

```
[centos@centosstream8 ~]$ sudo wc -l /var/lib/suricata/rules/suricata.rules
61502 /var/lib/suricata/rules/suricata.rules
```

17. Open the rules file and check the existence of the following detection rule.

Refer to the appendix and to the suricata rule webpage to understand each element of this rule:

nano /var/lib/suricata/rules/suricata.rules

```
  GNU nano 2.9.8       /var/lib/suricata/rules/suricata.rules

alert ip any any -> any any (msg:"SURICATA Applayer Mismatch protocol both $
alert ip any any -> any any (msg:"SURICATA Applayer Wrong direction first D$
alert ip any any -> any any (msg:"SURICATA Applayer Detect protocol only on$
alert ip any any -> any any (msg:"SURICATA Applayer Protocol detection skip$
alert tcp any any -> any any (msg:"SURICATA Applayer No TLS after STARTTLS"$
alert tcp any any -> any any (msg:"SURICATA Applayer Unexpected protocol"; $
alert pkthdr any any -> any any (msg:"SURICATA IPv4 packet too small"; deco$
alert pkthdr any any -> any any (msg:"SURICATA IPv4 header size too small";$
```

18. From the Virtual machine, generate a HTTP traffic to www.google.com website using the user-agent string "BlackSun"

yum install curl

```
[centos@centosstream8 ~]$ sudo yum install curl
Last metadata expiration check: 0:06:22 ago on Wed 15 Oct 2025 04:10:13 PM IST.
Package curl-7.61.1-22.el8.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package              Architecture  Version                 Repository     Size
================================================================================
Upgrading:
 curl                 x86_64        7.61.1-34.el8           baseos        353 k
 libcurl              x86_64        7.61.1-34.el8           baseos        304 k
 libssh               x86_64        0.9.6-14.el8            baseos        220 k
 libssh-config        noarch        0.9.6-14.el8            baseos         21 k
```

```
Upgraded:
  curl-7.61.1-34.el8.x86_64          libcurl-7.61.1-34.el8.x86_64
  libssh-0.9.6-14.el8.x86_64         libssh-config-0.9.6-14.el8.noarch

Complete!
```

curl -A "BlackSun" www.google.com

Indication: curl is a tool to transfer data from or to a server, using one of the supported protocols (FTP, FTPS, HTTP, HTTPS, IMAP, IMAPS, LDAP, LDAPS, POP3, POP3S, ...) The -A option specifies the User-Agent string to send to the HTTP server.

```
[centos@centosstream8 ~]$ curl -A "BlackSun" www.google.com
<!doctype html><html dir="rtl" itemscope="" itemtype="http://schema.org/WebPage"
 lang="ar-TN"><head><meta content="text/html; charset=UTF-8" http-equiv="Content
-Type"><meta content="/images/branding/googleg/1x/googleg_standard_color_128dp.p
ng" itemprop="image"><title>Google</title><script nonce="7BwvsC7DCYI5 vM9oLieXw"
```

19. The attempted connection should generate the following alert in /var/log/suricata/fast.log file

tail -f /var/log/suricata/fast.log

Terminal 2 :

```
[centos@centosstream8 ~]$ sudo tail -f /var/log/suricata/fast.log
10/15/2025-16:20:44.411978  [**] [1:2008983:8] ET USER_AGENTS Suspicious User Ag
ent (BlackSun) [**] [Classification: A Network Trojan was detected] [Priority: 1
] {TCP} 10.0.2.15:45514 -> 216.58.204.228:80
```

III. Testing Suricata response to malicious traffic

20. Check the existence of the following rule in the file /var/lib/suricata/rules/suricata.rules which serves in the detection of suspicious IP addresses

```
[centos@centosstream8 ~]$ sudo nano  /var/lib/suricata/rules/suricata.rules
```

21. On your VM, install hping3 tool to perform an SYN Flooding DoS attack.

yum install https://epel.mirror.liquidtelecom.com/epel/8/Everything/x86_64/Packages/h/hping3-0.0.20051105-33.el8.x86_64.rpm

```
[centos@centosstream8 ~]$ sudo yum install https://epel.mirror.liquidtelecom.com
/epel/8/Everything/x86_64/Packages/h/hping3-0.0.20051105-33.el8.x86_64.rpm
Last metadata expiration check: 0:44:31 ago on Wed 15 Oct 2025 04:10:13 PM IST.
hping3-0.0.20051105-33.el8.x86_64.rpm             46 kB/s | 105 kB     00:02
Package hping3-0.0.20051105-33.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

23. Generate an SYN Flooding DOS attack on a chosen target machine (The target should be in the same local network. You can choose your main host, or one of your neighbors' VMs).

hping3 -S -p 2020 --flood --rand-source 192.168.4.20

Refer to man hping3 to understand the meaning of every option in the executed command

Terminal 3 :

```
[centos@centosstream8 ~]$ sudo hping3 -S -p 2020 --flood --rand-source 192
.168.4.20
HPING 192.168.4.20 (enp0s3 192.168.4.20): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.4.20 hping statistic ---
1211907 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

24. While hping3 is running, tail the alert logs on Suricata to display the generated alerts

Terminal 2 :

```
[centos@centosstream8 ~]$ sudo tail -f /var/log/suricata/fast.log
10/18/2025-19:42:19.758640  [**] [1:2400002:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 3 [**] [Classification: Misc Attack] [Priority:
 2] {TCP} 42.136.192.67:27520 -> 192.168.4.20:2020
10/18/2025-19:42:19.976765  [**] [1:2400002:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 3 [**] [Classification: Misc Attack] [Priority:
 2] {TCP} 42.133.23.74:28071 -> 192.168.4.20:2020
10/18/2025-19:42:20.386505  [**] [1:2400019:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 20 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 112.142.11.96:29190 -> 192.168.4.20:2020
10/18/2025-19:42:20.397153  [**] [1:2400027:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 28 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 163.250.237.41:29144 -> 192.168.4.20:2020
10/18/2025-19:42:24.644367  [**] [1:2400024:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 25 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 147.119.146.90:29468 -> 192.168.4.20:2020
10/18/2025-19:42:24.854809  [**] [1:2400028:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 29 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 168.64.59.5:29801 -> 192.168.4.20:2020
10/18/2025-19:42:25.070887  [**] [1:2400030:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 31 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 175.103.119.252:30020 -> 192.168.4.20:2020
10/18/2025-19:42:25.132054  [**] [1:2400026:4495] ET DROP Spamhaus DROP Li
sted Traffic Inbound group 27 [**] [Classification: Misc Attack] [Priority
: 2] {TCP} 158.249.189.0:30169 -> 192.168.4.20:2020
```

Terminal 1 :

```
18/10/2025 -- 19:40:32 - <Info> - All AFP capture threads are running.
^C18/10/2025 -- 19:42:40 - <Notice> - Signal Received.  Stopping engine.
18/10/2025 -- 19:42:41 - <Info> - time elapsed 128.889s
18/10/2025 -- 19:42:42 - <Info> - Alerts: 52
18/10/2025 -- 19:42:45 - <Info> - cleaning up signature grouping structure
... complete
```

IV. Adding our own rules to Suricata

Detecting Echo Requests

In this step, we will add a rule to suricata to detect the use of icmp traffic. For that, you are asked to follow the next steps:

25. Create a rule file custom.rules under the directory /var/lib/suricata/rules/

vi /var/lib/suricata/rules/custom.rules

and add the following rule:

```
alert icmp $HOME_NET any -> $EXTERNAL_NET any (icode:0; itype:8; msg: "Ping Detected"; sid:1000001; rev:1;)
```

```
 GNU nano 2.9.8       /var/lib/suricata/rules/custom.rules

alert icmp $HOME_NET any -> $EXTERNAL_NET any (icode:0; itype:8; msg: "Pi$
```

26.
27.

28. Specify custom.rules as a rule file in the configuration file /etc/suricata/suricata.yaml

```
 GNU nano 2.9.8           /etc/suricata/suricata.yaml

## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
  - suricata.rules
  - custom.rules
```

29. Stop suricata (CTRL + C) and then restart it

suricata -c /etc/suricata/suricata.yaml -v -i enp0s3

```
[centos@centosstream8 ~]$ sudo suricata -c /etc/suricata/suricata.yaml -v
-i enp0s3
18/10/2025 -- 23:19:05 - <Notice> - This is Suricata version 6.0.20 RELEAS
E running in SYSTEM mode
18/10/2025 -- 23:19:05 - <Info> - CPUs/cores online: 2
18/10/2025 -- 23:19:05 - <Info> - Setting engine mode to IDS mode by defau
lt
```

30. Open a new terminal and ping to the external IP address, check the Suricata log file fast.log, and verify that your rule has been considered by your IDS.

ping 8.8.8.8

```
[centos@centosstream8 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=70.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=53.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=53.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=53.9 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 53.285/57.856/70.849/7.509 ms
```

tail -f /var/log/suricata/fast.log

```
[centos@centosstream8 ~]$ sudo tail -f /var/log/suricata/fast.log
10/18/2025-23:20:25.871957  [**] [1:1000001:1] Ping Detected [**] [Classif
ication: (null)] [Priority: 3] {ICMP} 10.0.2.15:8 -> 8.8.8.8:0
10/18/2025-23:20:26.873432  [**] [1:1000001:1] Ping Detected [**] [Classif
ication: (null)] [Priority: 3] {ICMP} 10.0.2.15:8 -> 8.8.8.8:0
10/18/2025-23:20:27.877766  [**] [1:1000001:1] Ping Detected [**] [Classif
ication: (null)] [Priority: 3] {ICMP} 10.0.2.15:8 -> 8.8.8.8:0
10/18/2025-23:20:28.878737  [**] [1:1000001:1] Ping Detected [**] [Classif
ication: (null)] [Priority: 3] {ICMP} 10.0.2.15:8 -> 8.8.8.8:0
```

Analyzing PCAP file and detecting phishing attacks

The objective is to analyze a pcap traffic capture, detect phishing attacks on HHTP and DNS queries, and write custom rules in SURICATA to detect such an attack.

31. Download the pcap file and open it using wireshark

wget https://github.com/jishnusaurav/Phishing-attack-PCAP-analysis-using-scapy/raw/master/phishingattack.pcap

```
[centos@centosstream8 ~]$ sudo wget https://github.com/jishnusaurav/Phishing-
attack-PCAP-analysis-using-scapy/raw/master/phishingattack.pcap
--2025-10-18 23:44:22--  https://github.com/jishnusaurav/Phishing-attack-PCAP
-analysis-using-scapy/raw/master/phishingattack.pcap
Resolving github.com (github.com)... 140.82.121.3, 64:ff9b::8c52:7903
Connecting to github.com (github.com)|140.82.121.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/jishnusaurav/Phishing-attack-PCAP
-analysis-using-scapy/master/phishingattack.pcap [following]
--2025-10-18 23:44:24--  https://raw.githubusercontent.com/jishnusaurav/Phish
ing-attack-PCAP-analysis-using-scapy/master/phishingattack.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.11
0.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.1
10.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 174149 (170K) [application/octet-stream]
Saving to: ‘phishingattack.pcap’

phishingattack.pca 100%[==============>] 170.07K   417KB/s    in 0.4s

2025-10-18 23:44:24 (417 KB/s) - ‘phishingattack.pcap’ saved [174149/174149]
```

dnf install wireshark -y

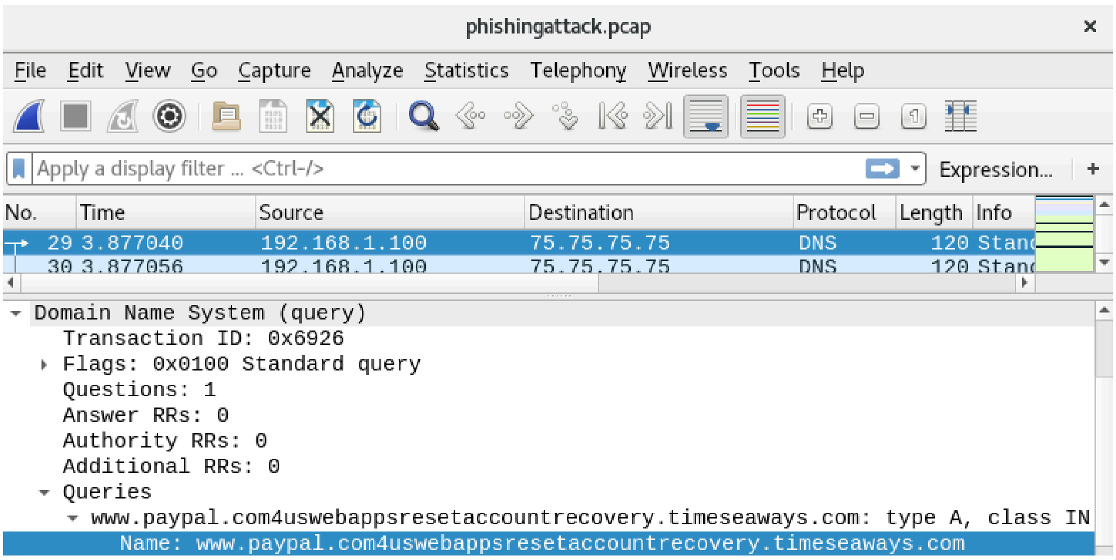
```
[centos@centosstream8 ~]$ sudo dnf install wireshark -y
Last metadata expiration check: 0:32:06 ago on Sat 18 Oct 2025 11:14:45 PM IS
T.
Dependencies resolved.
================================================================================
 Package                 Arch        Version                 Repository     Size
================================================================================
Installing:
 wireshark               x86_64      1:2.6.2-17.el8          appstream     3.6 M
Installing dependencies:
 libatomic               x86_64      8.5.0-22.el8            baseos         26 k
 libsmi                  x86_64      0.4.8-23.el8            appstream     2.4 M
 openal-soft             x86_64      1.18.2-7.el8            appstream     394 k

Complete!
```

32. Open packet 29, which represents a DNS query, and examine the content of the DNS protocol, and observe the following DNS query:

Open PCAP file with Wireshark (GUI)

```
[centos@centosstream8 ~]$ wireshark phishingattack.pcap
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayl
and to run on Wayland anyway.
```

packet 29 :
www.paypal.com4uswebappsresetaccountrecovery.timeseaways.com



Is it a phishing attack? If yes, how it works?

➔ Yes, it's a phishing attack. The attacker creates a fake domain starting with "paypal.com" to trick users into thinking they're on the real PayPal site, but the actual malicious domain is "timeseaways.com" where credentials are stolen.

33. To write a Suricata signature for this attack, we need to find and handle necessary patterns from the payload.

We can write a signature to detect any value after the real PayPal domain (paypal.com). For that, we use the “isdataat” relative keyword (See appendix for a better description of the keyword usage).

In the rule file custom.rules under the directory **/var/lib/suricata/rules/** add the following signature to detect the phishing attack.

```
alert dns $HOME_NET any -> $EXTERNAL_NET 53 (msg:"Paypal Phishing Detected";
dns_query; content:"paypal.com"; nocase; isdataat:1,relative; sid:1000003; rev:1;)
```

```
 GNU nano 2.9.8        /var/lib/suricata/rules/custom.rules

alert icmp $HOME_NET any -> $EXTERNAL_NET any (icode:0; itype:8; msg: "Ping $

alert dns $HOME_NET any -> $EXTERNAL_NET 53 (msg:"Paypal Phishing Detected";$
```

The traffic in phishingattack.pcap contains patterns related to other attacks.
In order to only test the above phishing attack signature and avoid generating other alerts when loading this pcap file, you can deactivate the rule file **suricata.rules** in the configuration file **/etc/suricata/suricata.yaml** (and only keep custom.rules active).

```
 GNU nano 2.9.8             /etc/suricata/suricata.yaml             Modified

##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
#  - suricata.rules
  - custom.rules
```

34. Stop suricata (CTRL + C) and then restart it

suricata -c /etc/suricata/suricata.yaml -v -i enp0s3

```
[centos@centosstream8 ~]$ sudo suricata -c /etc/suricata/suricata.yaml -v
-i enp0s3
19/10/2025 -- 00:16:56 - <Notice> - This is Suricata version 6.0.20 RELEAS
E running in SYSTEM mode
19/10/2025 -- 00:16:56 - <Info> - CPUs/cores online: 2
19/10/2025 -- 00:16:56 - <Info> - Setting engine mode to IDS mode by defau
lt
19/10/2025 -- 00:16:56 - <Info> - master exception-policy set to: auto
```

35. Open new terminal and replay the pcacp file (phishingattack.pcap) over the network interface enp0s3 using the command:

install tcpreplay :

```
[centos@centosstream8 ~]$ sudo dnf install tcpreplay -y
Last metadata expiration check: 1:05:04 ago on Sat 18 Oct 2025 11:14:45 PM
 IST.
Dependencies resolved.
============================================================================
 Package            Architecture   Version                Repository    Size
============================================================================
Installing:
 tcpreplay          x86_64         4.5.2-1.el8            epel         357 k
```

```
Installed:
  tcpreplay-4.5.2-1.el8.x86_64

Complete!
```

tcpreplay -i enp0s3 phishingattack.pcap

```
[centos@centosstream8 ~]$ sudo tcpreplay -i enp0s3 phishingattack.pcap
^C User interrupt...
sendpacket_abort
Actual: 436 packets (56314 bytes) sent in 102.81 seconds
Rated: 547.7 Bps, 0.004 Mbps, 4.24 pps
Flows: 65 flows, 0.63 fps, 430 unique flow packets, 6 unique non-flow packets
Statistics for network device: enp0s3
        Successful packets:        435
        Failed packets:            0
        Truncated packets:         0
        Retried packets (ENOBUFS): 0
        Retried packets (EAGAIN):  0
```

tcpreplay-edit -C -i enp0s3 phishingattack.pcap

```
[centos@centosstream8 ~]$ sudo tcpreplay-edit -C -i enp0s3 phishingattack.pcap
^C User interrupt...
sendpacket_abort
Actual: 173 packets (28716 bytes) sent in 16.30 seconds
Rated: 1761.5 Bps, 0.014 Mbps, 10.61 pps
Flows: 58 flows, 3.55 fps, 171 unique flow packets, 2 unique non-flow packets
Statistics for network device: enp0s3
        Successful packets:        172
        Failed packets:            0
        Truncated packets:         0
        Retried packets (ENOBUFS): 0
        Retried packets (EAGAIN):  0
```

36. Check the Suricata log file fast.log, and verify that the attack was successfully detected:

tail -f /var/log/suricata/fast.log

```
10/19/2025-00:27:44.758830  [**] [1:1000003:1] Paypal Phishing Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.1.100:47034 -> 75.75.75.75:53
10/19/2025-00:27:44.769173  [**] [1:1000003:1] Paypal Phishing Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.1.100:47034 -> 75.75.75.75:53
10/19/2025-00:27:49.780804  [**] [1:1000003:1] Paypal Phishing Detected [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.1.100:47034 -> 75.75.75.75:53
```